Angles of similar triangles

1

a $x = 28, y = 89$ **b** $a = 64, b = 85$ **c** $p = 71, q = 38$ **d** $r = 52, s = 38$

2 127°

Sides of similar triangles

3

a $m = 10, n = 21$ **b** $g = 30, f = 5$ **c** $x = 42, y = 20$ **d** $a = 9, b = 66$

4

a NL **b** NM

Scale factor

5

a 3 **b** 3 **c** $\frac{3}{2}$ **d** $\frac{5}{4}$

6

a i 2 ii $x = 8$

b i $\frac{8}{7}$ ii $a = 32$

c i 3 ii $x = 12$

d i 5 ii $y = 5$

e i $\frac{7}{2}$ ii $p = 10$

f i $\frac{8}{3}$ ii $q = 56$

g i $\frac{3}{2}$ ii $x = 12$

h i $\frac{2}{7}$ ii $m = 4$

7

a True **b** True **c** False **d** True

e True

8 4

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9 No solution provided.

10

a EF **b** $\angle EHG$

11

a 3 **b** $1:3$

12

a 0.5 **b** $2:1$

Applications

13 564 cm

14 23 m

15 $h = 15$

16 $h = 3.1$

17 $h = 1.4$

18 $h = 11.1$

19 $h = 4.8$

20 $d = 17.1$

21 $AB = 384\text{ m}$

22 $w = 77$

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